Introduction

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Abstract

This chapter introduces the biological and mathematical setting of the thesis. It motivates a stochastic treatment of early *Yersinia pestis* infection, reviews branching-process ideas used throughout, sketches the biphasic pathogenesis of plague, and situates the basic model of viral replication (BMVR) and bursting-versus-budding release strategies. Research questions and a roadmap of the remaining chapters are given at the end.

Contents

Yersinia pestis is the causative agent of bubonic plague. Perhaps more than any other human pathogen it has struck fear into the hearts and minds of peoples all over the world, and, consciously or not, it lingers as a shadow in the western psyche.

This thesis is about the mechanisms the bacterium employs inside a mammalian host, and about one of them in particular. *Y. pestis* does not resist the immune system from the outset. It permits itself to be swallowed by the host's own phagocytes, replicates inside them for a while, and only then reverses course and refuses uptake altogether. That reversal is among the sharpest differences between *Y. pestis* and the comparatively mild organism it descends from, and it is the sort of thing that invites a mathematical answer, because it is a choice between two strategies whose returns can in principle be written down and compared.

The picture drawn here is a stochastic one, and it is drawn at the scale where the choice is actually made: a few bacteria inside a single cell. What such a population does is settled by chance rather than by its average behaviour, so the object needed throughout is a distribution and not a trajectory. Most of the mathematics in this thesis is the machinery that requirement turned out to need.

1 Bubonic plague

Y. pestis was identified in 1894 by Alexandre Yersin, during the epidemic then running in Hong Kong. It was later attributed — and only recently confirmed with ancient genomic data — to a series of pandemics which killed a huge proportion of the European population and went on to shape the course of western cultural development.

The first of these opened with the Justinianic plague of 541, and its recurrences ran on into the eighth century. The second and most dramatic began in the 1340s and may have killed as much as half of the European population [1], a figure reaching above 80% in some urban centres. Successive waves of infection followed, and the continent did not fully emerge from the plight until as late as the seventeenth century, with the fire of London in 1666 often taken as the watershed moment.

There were likely earlier *Y. pestis* epidemics reaching far back into history, beyond the scope of accurate written account and recorded only in legend and myth. The organism is now thought to have emerged from its evolutionary antecedent, *Yersinia pseudotuberculosis*, some six thousand years ago, during what would have been a period of human population expansion driven by the development of agriculture. One may hypothesise that it was this excess supply of replicative means which provided the raw material for evolution, facilitating a shift of phenotype over the saddle and into a new local optimum — the niche pathogenesis observed in *Y. pestis* today. Whether rates of diversification really do move with the supply of room is a question taken up, in a quite different setting, by Chapter 9.

2 Stochastic modelling

Generally, at the outset of an infection, a relatively small number of colony forming units — bacteria or virions — will enter the body. Be it during inhalation into the respiratory system, through the skin with an insect's bite, or via ingestion, those initial pathogenic particles will be critical in determining the fate of the infection.

The founding population will either be depleted to extinction, or expand to a capacity where its growth becomes established and requires a more targeted effort of the immune system to deal with. Which of the two happens depends on causes of a chance nature: encounters with cells and other

(a) Bruegel, *The Triumph of Death*, c. 1562.

(b) Spadaro, *Piazza Mercatello* during the plague of 1656.

Figure 1: Plague in the European imagination, in allegory and in reportage. (a) *The Triumph of Death*, Pieter Bruegel the Elder, c. 1562 (Museo del Prado, Madrid). An army of skeletons lays waste to a landscape in which every human occupation is interrupted at once; in the lower right the card game, the banquet and the lute player have not yet noticed. The painting is an allegory of death in general rather than a record of an epidemic, but it was made in a Europe that had lived with recurrent plague for two centuries. (b) *Piazza Mercatello a Napoli durante la peste del 1656*, Domenico Gargiulo, called Micco Spadaro (Museo Nazionale di San Martino, Naples). Here there is no allegory: a single square in a real city during a real outbreak, the dead laid out in rows and the living still at work among them. Naples is thought to have lost something close to half its population in that year. Both public domain.

immune factors that are subject to the randomness of Brownian motion.

Stochastic effects play a more significant role in a population which is small. As the count of individuals increases, the dynamics of overall growth and decline become increasingly subject to the law of large numbers, and the expected fate of a particular bacterium becomes telling of the group's trajectory. Below that point a deterministic rate equation answers the wrong question. It returns a single trajectory, growing or decaying, when what one wants to know is whether the population establishes itself at all, and how large it is if it does. Both are questions about a distribution, and neither is visible in a mean.

For this reason, because we are interested in the early stages of a *Y. pestis* infection, it will be important to use stochastic models in our analysis.

3 Branching

Branching is one of the most fundamental patterns of behaviour observable in living systems. It is a basic means for the ordered structures of living entities to expand indefinitely in response to resource supply. Dividing one individual into two allows biomass, and hence dissipative potential, to increase without paying the cost required to increase agent complexity. At the most basic level, replication of DNA molecules makes up the foundational component of all biological life: the maintenance and development of information, held up and carried forward against all odds of decay.

Understanding branching from a mathematical perspective has been a topic of probability theory since the end of the nineteenth century. Darwin's cousin Francis Galton sought to understand the persistence of family names over generations, and for this, with Henry Watson, he employed what would come to be known as the Galton–Watson process [2]. It is a discrete-time stochastic process in which individuals either die or leave some number of offspring at each generation, fate being dictated by defined probabilities. In the binary case used throughout this thesis, each individual present either replicates with probability p or dies with the remaining likelihood, so that the number Z of offspring left by one individual has mean

$$\mathbb{E}(Z) = 2p + 0\,(1-p) = 2p.$$

It wasn't until the 1920s that continuous time models of branching were developed. In 1924, George Udny Yule published a paper in response to a book on evolution. The book was called "Age and area" [3], and sought to understand the relationship between the number of species in a living genus and the area over which it spans. The paper [4] introduced what would become known as the Yule process. This was the first continuous time model of branching. Individuals only replicate, and do so with some probability during an interval of time:

$$\Pr\left\{\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + 1 \;\mid\; \mathbf{X}_t\right\} = \lambda \mathbf{X}_t \Delta t + o(\Delta t).$$

As the interval is taken in the limit, $\Delta t \to 0$, the probability that more than one replication takes place becomes infinitesimal. The fact that any of the $\mathbf{X}_t$ present individuals can reproduce is encoded by the proportionality of the birth rate $\lambda \mathbf{X}_t$.

This model, applied equally well to bacterial replication as to the emergence of species, is also known as the standard birth process. Introducing death events, we get the standard birth–death process:

$$\Pr\left\{\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + 1 \;\mid\; \mathbf{X}_t\right\} = \lambda \mathbf{X}_t \Delta t + o(\Delta t)$$

$$\Pr\{\mathbf{X}_{t+\Delta t} = \mathbf{X}_t - 1 \mid \mathbf{X}_t\} = \mu \mathbf{X}_t \Delta t + o(\Delta t),$$

which now has a universally accessible absorbing state at 0 representing extinction.

Criticality

One of the most important aspects of branching processes is their net replacement, which asks whether an individual replaces itself on average. In discrete time this is the expected number of offspring produced by a single individual over its lifetime, $\mathbb{E}(Z) = 2p$, and the process is called **supercritical**, **critical** or **subcritical** according as $2p$ exceeds, equals or falls below 1. In continuous time the same question is asked of the mean population size, which obeys

$$\frac{\mathrm{d}\mathbb{E}(\mathbf{X}_t)}{\mathrm{d}t} = (\lambda - \mu)\mathbb{E}(\mathbf{X}_t),$$

so that the three regimes are set instead by the sign of the net replication rate $\lambda - \mu$. The two conditions say the same thing in the two settings, and the distinction between them is one of arithmetic rather than of substance. The three regimes behave quite differently. Some supercritical realisations die out and the rest rise towards infinity in an increasingly predictable exponential fashion. Every subcritical realisation is destined to fixate at 0. So is every critical one — but the expected time it takes to do so is infinite, so that a single realisation may last a very long time without ever fixating.

Catastrophes

Much of the work presented in this thesis surrounds our analysis of a further modification to this birth–death process. An additional "catastrophe event" is introduced, whereby a sudden transition can take place from any non-zero count either to 0 or to a separate ruptured state H, as per choice of convention:

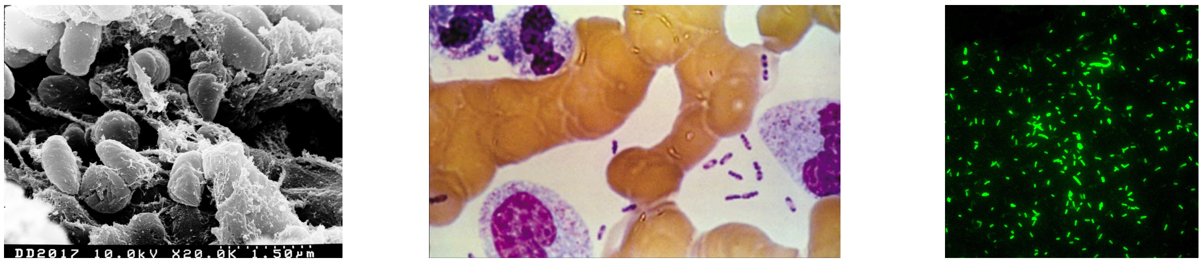
$$\begin{aligned}
\Pr\{\mathbf{X}_{t+\Delta t} = \mathbf{X}_t + 1 \mid \mathbf{X}_t\} &= \lambda \mathbf{X}_t \Delta t + o(\Delta t) \\
\Pr\{\mathbf{X}_{t+\Delta t} = \mathbf{X}_t - 1 \mid \mathbf{X}_t\} &= \mu \mathbf{X}_t \Delta t + o(\Delta t) \\
\Pr\{\mathbf{X}_{t+\Delta t} = H \mid \mathbf{X}_t\} &= \delta \mathbf{X}_t \Delta t + o(\Delta t).
\end{aligned}$$

Like birth and death, the catastrophe can be initiated by any of the individual units, and so occurs at a rate proportional to the population size. In our case this is to represent the possibility of any resident bacterium within a white blood cell triggering a bursting event in which the walls of the macrophage disintegrate and the pathogenic contents are simultaneously released.

The birth–death–catastrophe process is the spine of this thesis. Chapter 4 and Chapter 5 solve the single cell; Chapter 6 carries it up to a population of cells; Chapter 7 solves a two-type version of it; and Chapter 8 asks what the burst is worth against a defence that can be used up.

4 Early phase *Y. pestis* infection

Y. pestis differs from its evolutionary antecedent, *Y. pseudotuberculosis*, in a few ways, but one stands out. *Y. pestis* exhibits two phases in its pathogenesis of mammalian hosts. In the first, the bacteria are readily absorbed by host leucocytes via phagocytosis — white blood cells, mainly macrophages and neutrophils — and within macrophages especially they replicate in relative shelter and secrecy from patrolling immune agents. In the second, after a switching period, they prevent this absorption by a combination of shielding and active interference. This work explores the mathematical implications of the switching in an attempt to quantify the significance of the strategy's evolutionary advantage.



(a) In the flea foregut. (b) Wayson stain. (c) Antibody stain.

Figure 2: The organism. (a) Scanning electron micrograph of *Y. pestis* massed in the foregut of a flea, where the bacteria form the biofilm blockage that makes the vector infectious; the scale bar in the original reads 1.5 μm. (b) Wayson stain, showing the bipolar "safety pin" appearance by which the bacillus is recognised, scattered among host leucocytes. (c) Direct fluorescent antibody stain, in which the antibody binds the F1 capsular antigen — one of the phase-two factors described below, and absent at flea temperature; the field gives a fair impression of the numbers involved. Sources: (a) NIAID; (b) and (c) CDC Public Health Image Library. All three are in the public domain.

Phase one: an unarmed inoculum, and a permissive compartment

The bacterium that arrives in a mammalian host is not the bacterium that later dominates it. In the classical bubonic cycle *Y. pestis* multiplies in the flea midgut at roughly ambient temperature, and at that temperature it produces little or none of the F1 capsule, while much of the temperature-induced mammalian virulence package is not fully on [5]. What a flea bite delivers into the dermis is therefore a vector-adapted population, unarmed for the host it has just entered. Most bacilli that encounter neutrophils at the bite site are killed; a minority are phagocytosed by tissue macrophages, which fail to sterilise the infection and instead supply a protected niche.

Inside the macrophage the bacteria occupy a specialised compartment, the *Yersinia*-containing vacuole. Rather than being delivered efficiently to a fully bactericidal phagolysosome, this vacuole remains spacious and relatively permissive, with acidification and autophagic killing both limited [6, 7]. Within it the organism multiplies and experiences a suite of host-associated stresses — body temperature, reduced pH, oxidative pressure, restricted metal availability. These do more than select for hardy cells; they act as signals which rewire gene expression, and the intracellular interval buys the time needed to synthesise the machinery that will define the second phase. The macrophage is as much a developmental window as a refuge.

Phase two: a surface that refuses re-entry

Temperature is the master cue. A shift from about 26–28°C to 37°C induces the F1 capsule, components of the type III secretion system, and many other virulence loci besides; the mildly acidic pH encountered in the vacuole specifically favours the pH 6 antigen [5]. Three classes of factor matter most. The F1 capsule, assembled as linear fibres over the cell surface, interferes with uptake and sterically masks the shorter adhesins that promoted entry in the first place. The pH 6 antigen promotes attachment in a way that does not efficiently trigger internalisation. And the Ysc–Yop type III secretion system, on contact with a host cell, injects effector proteins directly into the eukaryotic cytosol: YopH disrupts the focal-adhesion signalling required for phagocytic uptake, while YopE, YopT and YpkA between them disable the Rho-family signalling that organises the actin cytoskeleton. Together these paralyse phagocytosis. The same cell that earlier needed invasive factors to enter a macrophage now presents a surface optimised to refuse re-entry.

Why not begin in phase two?

If the anti-phagocytic state is what enables explosive extracellular growth, why does the organism not arrive already in it? The literature gives several reasons, and they are not exclusive [5, 6]. Under natural transmission phase two is simply unavailable at the moment of inoculation, because the flea is cool and the programme is temperature-cued. Even once the cue is received, reprogramming is not instantaneous — capsule assembly and build-up of the secretion apparatus take hours — and neutrophils do not wait. The two surfaces are partly antagonistic, so a bacterium presenting a full phase-two coat would impair the very macrophage entry which, under natural transmission, supplies the window in which to adapt. And a permanently assembled secretion system is metabolically expensive, and of no use at all in the flea.

Those are reasons why the sequence is *forced*. They leave open the separate question this thesis is about: granted that the sequence is forced, is it also worth something? Does passing through a protected compartment and arriving outside in numbers improve the odds of establishment by enough to matter, and if so, by how much?

The sequence in outline

At the outset of invasion, colony forming units are phagocytosed by host macrophages, and with evolved resistance to their toxins the bacteria use this intracellular refuge as a temporary home in which to replicate and bolster numbers against challenges to come. The first mounted attack of the innate immune system is neutrophils. Summoned by various cytokines, they arrive on the scene in numbers, passing through the walls of blood vessels into the extracellular matrix in which the infection is beginning to take hold. After some time an infected macrophage will rupture and its resident bacterial population will disperse. Those *Y. pestis* bacteria which have undergone the transition to the second phase are characterised by being resistant to further phagocytosis. Figure 4 illustrates the dynamics.

The exit, which is not settled

Teaching summaries put the last step of that sequence bluntly: the bacteria replicate inside macrophages, the macrophages burst, and free bacilli spread. The primary literature is more careful. Phase-two machinery certainly can kill macrophages — YopJ-dependent apoptosis is well supported [8], in activated macrophages the infection can be redirected towards caspase-1-dependent pyroptosis, and necroptosis of infiltrated macrophages has been proposed to promote dissemination in lymph nodes [9] — but the evidence that this is how intracellular *Y. pestis* actually exits after the first phase is weaker and incomplete [10]. The exit route remains an open problem.

That gap matters here more than it might elsewhere. The catastrophe rate δ of the previous section is precisely the parameter the exit mechanism fixes, and it is the one number that would place *Y. pestis* on the scales built in Chapter 6 and Chapter 8. Nothing in this thesis is fitted to plague data, and Chapter 10 says plainly what would have to be measured for that to change.

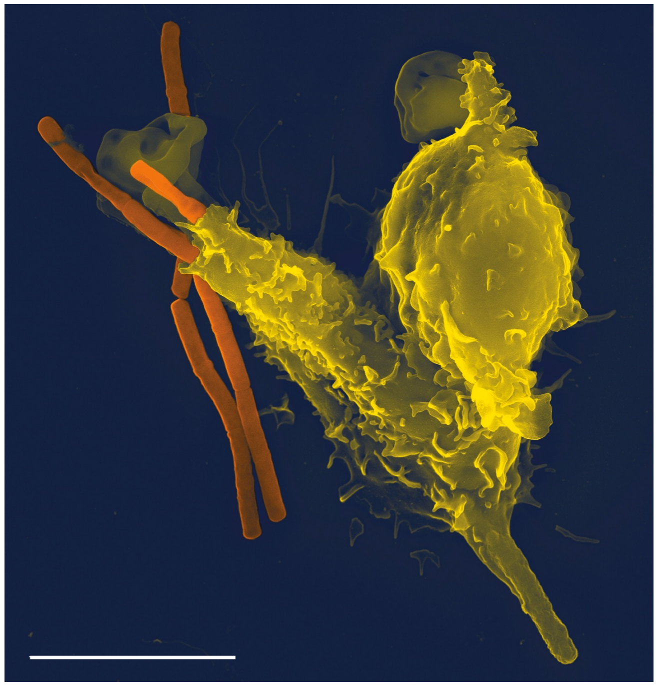


Figure 3: Phagocytosis, the act on which the whole strategy turns. A neutrophil (yellow) engulfing bacilli (orange); the scale bar is $5\,\mu$m. This is what *Y. pestis* permits in its first phase and refuses in its second, and it is what the injected effectors YopH, YopE, YopT and YpkA exist to prevent. The bacteria shown are *Bacillus anthracis* rather than *Y. pestis*. Image by Volker Brinkmann, cover of *PLoS Pathogens* 1(3), 2005, reproduced under CC BY 2.5.

5 HIV and the basic model of viral replication

In the 1980s there was a major global HIV epidemic. This was terrible, and the disease has killed on the order of forty million people since, but it had the side effect of inducing a flurry of research into HIV pathogenesis and viral dynamics more generally. By the early 1990s progress was being made with the development of effective mathematical models describing within host viral replication.

Principal among these was what became known as the Basic Model of Viral Replication (BMVR) [11, 12, 13], which emerged from the fray of model diversity following a period of substantial experimentation. Its basic assumption was quite simple: there are three agent types.

- **Target cells:** T

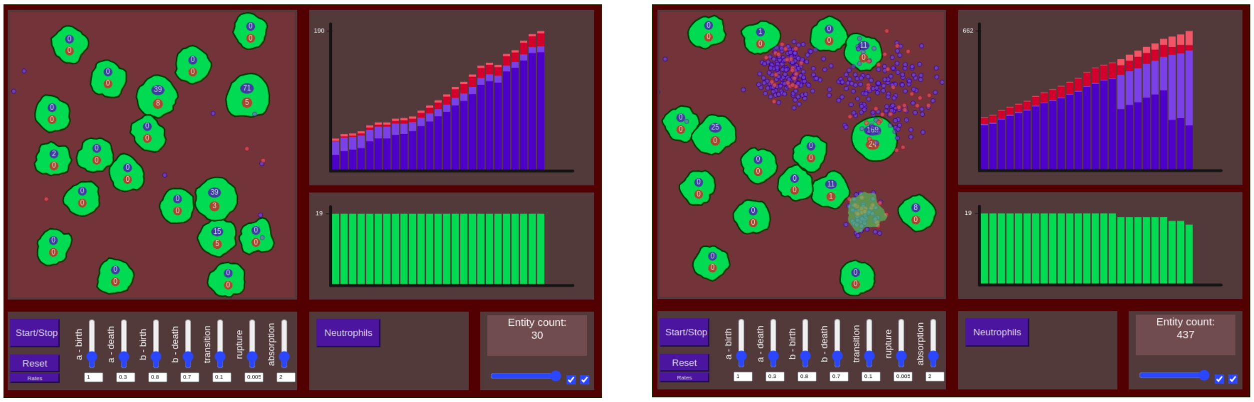
  Host cells produced with a constant influx of rate s, removed as they become infected by individual virions with a rate γ per cell–virion pair.

- **Free virions:** V

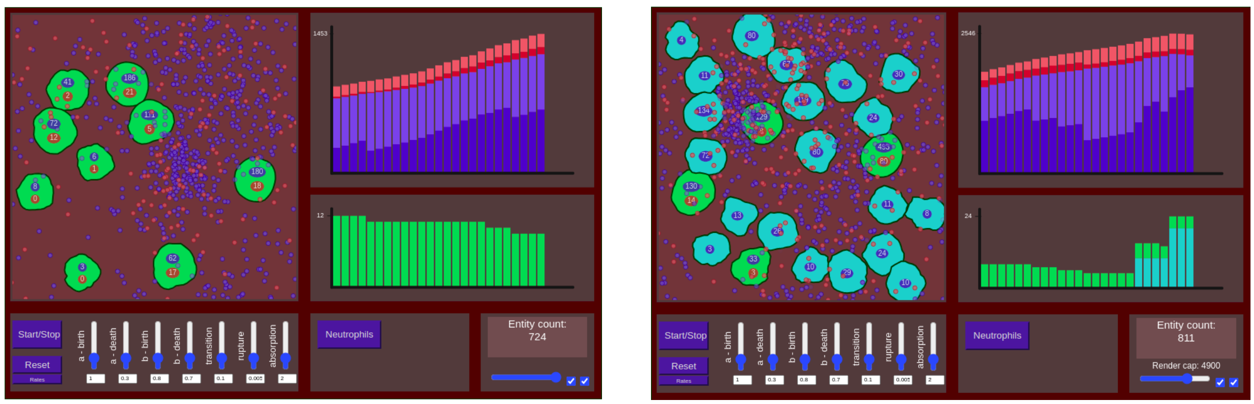
  Produced at a constant rate, p, by each infected cell, and also cleared with rate c.

- **Infected cells:** I

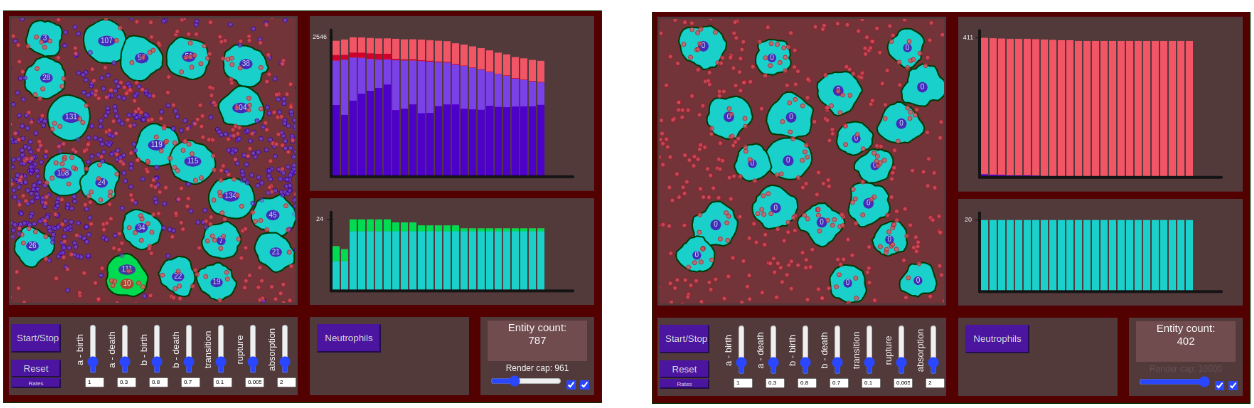
  A single free virion coming into contact with a target cell is sufficient to convert it into an infected cell, which removes one cell from T. Infected cells then produce new free virions at a constant rate, not accounting for the likely reality that a higher intracellular load might make a cell more productive, though some models do include a lag period between the time a cell is



Bacterial replication takes place within the macrophage. Some phase transitions occur.



The macrophage bursts; more transitions to the second phase (red particles). Neutrophils arrive.



Neutrophils phagocytose and eliminate the remaining phase one bacteria, leaving *Y. pestis* as a primarily extracellular infection.

Figure 4: A schematic representation.

infected and when it begins to produce virions. They are also removed by cell death with rate d_I.

The equations are as follows,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = s - \gamma TV$$
$$\frac{\mathrm{d}I}{\mathrm{d}t} = \gamma TV - d_I I$$
$$\frac{\mathrm{d}V}{\mathrm{d}t} = pI - cV.$$

In some cases the model is simplified by taking the number of target cells as a constant, T. This represents the situation in which there is a large availability of cells to infect, such that their depletion is not a significant driver of the overall dynamics. With this simplifying assumption the equations become

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \gamma TV - d_I I$$
$$\frac{\mathrm{d}V}{\mathrm{d}t} = pI - cV.$$

This is the form carried through Chapter 6. To be clear, it rests on the assumption that virions are produced by cells one by one at a constant release rate, and that release and cell death are separate, independent events.

New research into HIV viral release

In 2019 an experiment was published [14] with the intention of quantifying the release dynamics of HIV virions from infected cells. Small wells were each seeded with a single infected cell, and viral counts were then measured from the liquid of every well at intervals over a period of days.

If the assumption of the BMVR were correct, we would expect to see something approximating a Poisson process in each well, with roughly the same count recorded in all of the wells at each interval. What was actually observed was very different: huge variation between wells, and production rates that were not at all constant. Instead, a high degree of intermittency was recorded — cells remaining completely inactive for successive intervals, then occasionally releasing a substantial number of virions, seemingly all at once. This challenges the standard assumption of constant-rate budding directly, at the level of the single cell where the assumption is made.

So what?

The observation begs the question: why? There will be some reason HIV has evolved to use this quasi-bursting procedure. Perhaps it is simply more convenient for release to take place in floods. Passage through the cell wall requires an opening, and it surely makes sense to induce this only periodically, avoiding both unwanted leakage of cellular components and ingress of extracellular debris.

But perhaps there is also some general advantage to be gained by virions being released in large cohorts like this. It could be that locally depletable immune factors are flooded and overcome by the sudden outflux of enemy agents. A paper by N. Komarova [15] presented this idea of a so-called "flooding" advantage, and it has an obvious analogy: a gang surrounded in a house may reason that if they all run out at once, some will get away in the confusion. Whether the analogy survives contact with a mathematical model is the subject of Chapter 8, with attention paid to the spectrum between budding release and complete cell bursting.

Back on the topic of the BMVR and its questionable assumption, the two constants p and d_I are phenomenological. They are fitted to viral load data and then reported as though they characterised the pathogen, when in fact nothing in the model says what the cell is doing between infection and release. For a budding virus that costs nothing, because release and death really are independent events. For a lytic pathogen it is wrong in mechanism: such a pathogen releases nothing while it replicates and then releases everything at once, in the event that ends the cell's life. Release and death are not independent; they are the same event. Chapter 6 replaces the two constants with two functions of cell age derived from the birth–death–catastrophe process, and recovers the BMVR as a projection of the result.

6 Pathogenic release strategies

With both viral and bacterial particles being very small, it has historically been extremely challenging to make good observations of their behaviour. Progress is taking place, and in the last couple of decades we have been able to observe individual bacteria through the course of an intracellular infection. Dstl, the Defence Science and Technology Laboratory, has been able to record videos of a macrophage bursting to release on the order of a hundred *F. tularensis* bacteria; through the induction of bacterial luminescence, each individual pathogenic particle can be visually isolated and counted. Stochastic models of that intracellular phase have been fitted to data of this kind [16].

F. tularensis and *Y. pestis* are similar in size, each around 0.5 micrometres (*Y. pestis* being marginally larger). Viruses, though, are generally about an order of magnitude smaller than bacteria, with HIV at approximately 100 nanometres. They are hence somewhat more elusive to visualisation and counting.

Bursting

Both *F. tularensis* and *Y. pestis* multiply within macrophages before causing the cell to lyse, releasing their pathogenic cohort for further replicative endeavours. The case of *Y. pestis* is slightly more complicated, with the biphasic switch from intracellular to extracellular replication described above. But generally, strategies which involve a complete release in combination with the death of the host cell are referred to as bursting pathogenesis.

Budding

As described above, budding release refers to the situation where an infected cell outputs new virions at a constant rate. Models of viral replication generally take this as an assumption, and it has become a commonly held belief that this is how viral release takes place.

Something in between?

The recent work on HIV release reveals that the story may not be so simple as was previously assumed — that in cases of non-lytic virulence there may be various strategies we are not fully aware of. Another example of something in between bursting and budding is release by extracellular vesicles. Among viruses that leave the cell non-lytically, a subdivision may be made into those which produce and assemble their own protein shell to harbour genetic material, and those which

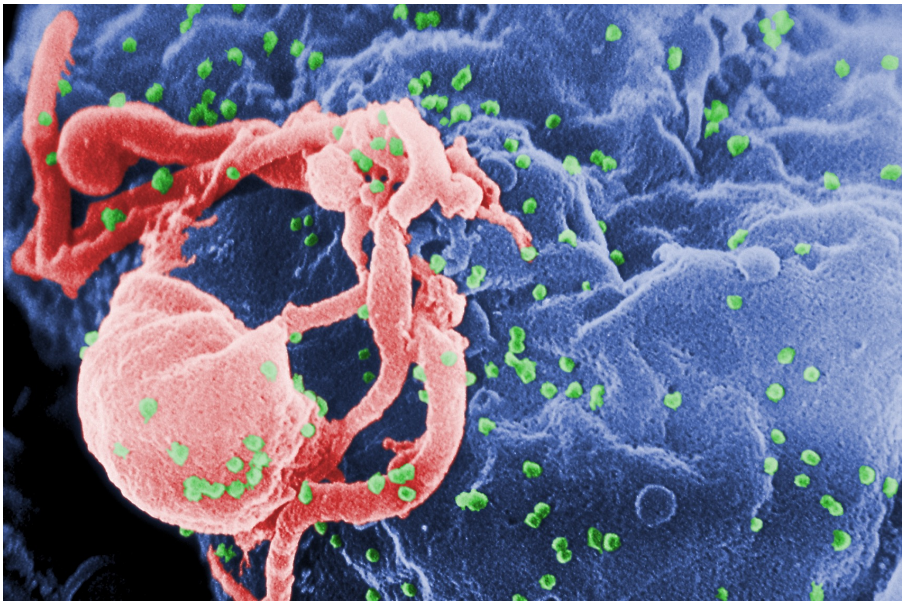


Figure 5: Budding release. Scanning electron micrograph of HIV-1 virions (green) assembling and departing from the surface of a cultured lymphocyte, each bump a separate event and the cell intact. Release and death are independent here, which is why the standard model's two constants describe such a cell exactly. A bursting pathogen does the opposite: nothing leaves while the load accumulates, and then everything leaves at once, in the event that ends the cell — and no pair of constants describes that. Chapter 6 turns on the difference. Image: CDC Public Health Image Library #10000, photo credit C. Goldsmith; public domain.

exploit the cell membrane to produce a lipid container. Of this second group it has been known for a while that in some cases multiple individual viruses are contained and released in one ex-membrane capsule; these are referred to as extracellular vesicles [17, 18], and they present another potential complexity to the story of viral release.

What decides whether any of this matters is not the schedule by itself. At fixed total output per cell and a defence that removes free particles at a constant rate, bursting and budding give exactly the same reproduction number; the schedule becomes visible only when the defence can be used up, or when the quantity of interest is the chance of establishment rather than a mean. Chapter 6 and Chapter 8 make that statement precise, and find that the comparison reverses according to whether the defence facing a release is a rate or a finite budget.

7 Research questions

There are four key areas of focus that this research seeks to address. The first two are questions into the nature of biological process.

- **Why did *Y. pestis* evolve a biphasic pathogenesis in mammalian hosts?** Switching from a pro-phagocytic to an anti-phagocytic strategy seems to be crucial to the overall pathogenesis of *Y. pestis*, and the preceding sections give reasons why the sequence is forced without saying what it is worth. There is a simple argument based on the net proliferability both before and after neutrophils arrive on the scene of infection, which sorts the bacterium's options into a two-by-two and locates a threshold neutrophil fraction at which the better option changes; it is set out in Chapter 10. A second argument, of a quite different kind, keys the same switch to the amount of killing the host can afford per delivery. And a third strand supports both without deciding either: if *Y. pestis* replication is only slightly supercritical, then a near-critical branching process becomes drastically less likely to die out when its initial count is increased by even a few colony forming units (Section 2.A). If we see the within-macrophage replication as setting up an initial population, this effect offers further explanatory potential.

- **Is there a general advantage to "bursting" style transfer in successive intracellular infection?** N. Komarova proposed that there could be a flooding advantage to bursting pathogenesis — bacteria or virus exiting their host cells all at once in large groups rather than one at a time or in dribs and drabs. Chapter 6 and Chapter 8 give mathematical support to the theory and, more usefully, say when it holds and when it does not.

The second two are areas in which we tried to construct new models to describe particular dynamics.

- **An intracellular replacement for the standard model of viral replication.** The two constants of the BMVR describe a budding cell exactly and a bursting cell not at all, since for a lytic pathogen release and death are the same event rather than independent ones. Based on our analysis of the birth–death–catastrophe process, those constants can be replaced by quantities derived from the intracellular dynamics, and the question is what that costs and what it buys.
- **Efforts to describe variable rates of evolution with Markov chains.** The standard birth–death process has been used for decades to model the emergence of species over geological time periods. It may however be criticised for being consistent with the outdated doctrine of progressive gradualism, which holds that evolutionary change is constant and happens gradually. In 1972 Eldredge and Gould proposed the counter theory of punctuated equilibria [19]: evolution taking place rapidly over proportionally short periods, intervened by longer times of relative morphological stasis. There is also the controversial question of why large clades appear generally to have experienced an initial period of substantial diversification, which some attribute to a statistical consequence of standard birth–death dynamics — the push of the past, and the pull of the present. This work presents models which seek to give mathematical basis to the punctuated-equilibria paradigm, and asks more generally what survives when a rate is allowed to move. There is potential significance here, too, in understanding the emergence of *Y. pestis* itself.

Chapter 10 returns to all four and answers them as far as the work allows, which in two cases is not all the way.

8 How the content is divided amongst the chapters

Broadly, the thesis divides into three parts. The first is groundwork: the stochastic processes and the methods used to solve them, together with one constant that took long enough to pin down to deserve a chapter of its own. The second is the birth–death–catastrophe process — the single infected cell, the population of cells built from it, a two-type version, and what the whole construction says about bursting pathogenesis. The third lets the rates move, and is where the mathematics of the earlier chapters gives out. Including this introduction and the conclusion, these parts are divided amongst ten chapters. Below is a summary of the content and message of each.

Chapter 2: Mathematical introduction

Much of the work presented in this thesis surrounds topics of stochastic process theory, and specifically branching processes. This chapter assembles the objects and the methods: Markov chains in discrete and continuous time, the branching processes that matter here, and the generating-function machinery — forward and backward equations, means and higher moments, hitting and extinction probabilities, limiting conditional means, and the method of characteristics — each developed once, on whichever process shows it most plainly. Two extensions then leave the time-homogeneous setting, in anticipation of Chapter 9: processes whose rates vary with time, and coupled systems in which a differential equation and a Markov chain each drive the other.

Two of its appendices are more than review. The first records how heavily survival near criticality depends on the size of the founding cohort (Section 2.A), which is the fact that almost everything downstream leans on. The second solves a family of absorption models, in which particles are drawn into a compartment and may then replicate inside it; the hardest member of that family is solved here for the first time (Section 2.B).

Chapter 3: Galton–Watson branching and the constant $A(p)$

For the subcritical binary Galton–Watson process the survival probability decays as $S_n \sim A(p)(2p)^n$, and that single constant also fixes the population conditioned on survival, whose mean converges to $1/A(p)$. The constant exists because the limit can be rewritten as a convergent infinite product. Rescaling the survival recursion to the logistic map identifies it with a value of the associated Koenigs linearising function, and a theorem of Becker and Bergweiler shows that function to be hypertranscendental [20]. That does not settle the closed-form question for $p \mapsto A(p)$, but it does explain why a long search for a formula came to nothing. What remains is practical: the product, together with the near-critical asymptotic $A(p) \sim 2(1-2p)$, describes the constant across the whole subcritical range.

This is the chapter furthest from plague, and it is kept because the question it answers is a real one and the answer is complete.

Chapter 4: The birth–death–catastrophe process

Around the 1980s, work was being done into growth models that incorporated stochastic declines. Out of this came the birth–death–catastrophe model as we know it [21, 22, 23, 24]: an extension of the standard birth–death process in which, from every state apart from 0, a catastrophe event can take place. We were interested in this model for its potential application to the development of an infection within cells that burst.

This chapter defines the process and derives the single-cell theory. The calculation closes because the probability of no rupture yet, $I(t)$, obeys an autonomous Riccati equation; the two roots a and b of its characteristic quadratic then carry every closed form in the thesis. Cell-fate probabilities, the mean and second moment of the intracellular load, the mean release and the first two conditional moments of the burst size all follow in that one coordinate. Some of the results have been found by others before, but several — the second moment expressed through I, the variances, and the joint load–release moments — go beyond what the existing literature had recorded.

Chapter 5: Distribution theory, quasi-stationarity and burst statistics

The previous chapter gives the moments; this one gives the laws. The intracellular load is geometric at every time, with a ratio that slides towards $1/a$; the quasi-stationary law — the behaviour of the process count given that fixation has not yet occurred — is that endpoint; and the burst size, averaged over the instant of rupture, is the same law again.

That the two coincide is the most satisfying result in the thesis, because the two constructions share no step: one conditions a surviving trajectory on never being absorbed, the other averages over rupture times, and the identity holds because size-biasing telescopes the occupation integral onto the endpoint of the process's own geometric ratio. Two probes then mark the boundary of the structure. Several founders sharing a cell leave it intact; chaining one cell's burst into the next destroys it as soon as internal extinction is possible, and that case remains open.

Chapter 6: Burst-aware within-host dynamics

This chapter embeds the cell in a population. The survival and release statistics of the previous two chapters become kernels indexed by cell age, and the resulting renewal system is exact in the mean rather than an approximation to it. Three consequences follow. The standard model of within-host virus dynamics is a projection of that system, valid one exponential phase at a time, so a fitted release rate is a property of the phase and not of the organism, and a model that already carries a hand-fitted eclipse alongside a bursting pathogen is fitting the same delay twice. No fit of the two population constants recovers the three intracellular rates, so a population fit is worth doing only alongside a single-cell measurement. And at fixed total output under linear clearance, the release schedule is invisible to R_0 while remaining visible to establishment: bursting strictly lowers the extinction probability of a small inoculum if and only if the flooding parameter $L = a(1 - b)$ exceeds one.

Chapter 7: Type-specific catastrophe rates

In 2011 Antal and Krapivsky [25] published an exact solution to the two-type continuous time birth–death process with one-way mutation. We were able to use their method, with some changes of variable, to obtain the equivalent result once catastrophes are allowed.

Type 1 converts irreversibly into type 2, either type may trigger a single global catastrophe, and because the catastrophe rate depends on the whole path, the natural object is an occupation-time transform rather than a terminal-count generating function. The backward system is triangular, and after linearisation the surviving Riccati equation becomes the Gauss hypergeometric equation, giving a closed formula for the finite-time probability of avoiding catastrophe which holds for any ordering of the type-specific rates. Read biologically, the two types are the early and adapted intracellular phenotypes of *Y. pestis* described above, and the catastrophe is irreversible loss of macrophage containment.

Chapter 8: Bursting and budding pathogenesis

Work has been done by others [15, 26, 27, 28] comparing the dynamical differences between budding and bursting pathogenesis. This chapter asks what a burst is worth when the defence facing it can be used up — when there is, as it were, a safety in numbers.

Inside the cell, a replicator–suppressor process treats the toxin-containing granules of a macrophage or neutrophil as a store which is spent and never replenished. A larger incident cohort is then disproportionately safer, because the cohort itself depletes the store. Outside it, in the extracellular medium, the pathogen meets complement proteins and killer cells which are likewise finite on the relevant timescale; modelling this as a fixed removal budget ϑ per delivery gives the surviving release in closed form, $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, so that every unit of removal costs the burst one factor of a. With it comes a reversal: against a threshold rather than a rate, the flooding criterion of the previous chapters flips, and bursting beats matched-mean budding precisely when $L < 1$. The reason is a single fact about variance — extinction probability rewards low offspring variance, whereas the surplus above a threshold is convex and rewards high variance — so the sign of $L - 1$ drives both criteria, and drives them opposite ways.

Chapter 9: Models with non-constant rates

Every model carried through to a result in the preceding chapters holds its rates fixed, and that assumption is what makes the generating functions close. It is also the first thing an ecologist will question. This chapter asks what happens when a rate is allowed to move, and it came from a different appetite from the rest of the thesis; it marks plainly which of its parts are results and which are specifications.

In the spirit of Yule's paper and of more recent work with birth–death processes [29], the first model gives a genus an inhomogeneous origination rate, driven by the room a population has left to expand into under logistic growth. The species count stays geometric and depends on the rate law only through its integral; under the logistic law that integral converges, leaving a proper limit law rather than merely a finite late-time mean. The second treats Pimentel's fly experiment [30, 31], in which two species were shown to coexist in one niche by oscillation, because a population pushed towards extinction evolves in response to interspecific pressure more than intraspecific; here the state at which the chain is equally likely to step up as down is available in closed form, and it is unstable, which is a different thing from a process approaching an equilibrium. Two further models, in which a population accumulates a potential and then loses it in a catastrophe, are carried only as far as simulation.

The chapter's general observation is worth repeating, because it is not the obvious one: what costs the generating function is not feedback as such but continuous bending of a rate. A process may act on its own circumstances as much as it likes and stay solvable, provided what survives the interaction is a random time or a random count rather than a moving rate.

Chapter 10: Conclusion

The conclusion is a reckoning rather than a summary. It sets out what each chapter established and what it left standing; returns to the four questions posed above and answers them as far as the work allows; gives the quadrant argument, a small piece of reasoning about why an organism might want to be eaten early and not late, which never found a home in a chapter of its own; and says plainly what the thesis does not establish, together with the four open problems worth taking first.

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